
Genre Based Music Generation using Deep Generative Models

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1 Research Problem

The goal of this project is to develop a deep generative model which serves as a genre-conditioned audio generation tool, able to create original audio content within a user-specified genre. The genre works as a creative constraint in the generation process, enabling the model to produce unique audio samples with the stylistic traits of the genre. These features include the rhythm, structure, instrumentation, pacing, and other characteristics that distinguish a genre. By learning the relationship between a genre and its audio representations, the model will capture the connection between low-level stylistic features and high-level overall structure. The output should reflect the model’s understanding of each genre’s characteristic sound. This project will explore the ability of deep generative models to abstract musical features.

2 Importance of this Problem

The relevance of this problem shows up in its focus on control in deep generative models, rather than purely random generation. By allowing users to select a genre, the model provides direct guidance over the generation process, which is a core objective in many applied generative systems where user intent must guide generation, like domain-specific LLMs or interactive image generation models. Past the technical importance, this solution is also valuable in terms of its practical uses. While this model likely won’t replace the traditional audio generation pipeline in the music industry, it has applications in areas where music is a secondary aspect of the final product, such as video games or social media. This solution would also improve the accessibility of music generation, as it allows non-musicians to create content without technical or musical experience.

3 Related Works and their limitations

One of the main works in audio generation is WaveNet, which shows that autoregressive models are able to work directly with raw audio waves. But, because it uses sequential sampling, generation is slow and computationally expensive. It is also limited in the control the user has to specify audio style or structure. When we look past the foundational works and into more recent works, such as Jukebox, we see that autoregressive modeling conditioned on rich user-provided inputs can guide music generation. However, this conditioning is reliant on a complex and large-scale pipeline, making simple genre only control difficult to isolate and also making it inefficient for lightweight applications. This infrastructure requires massive datasets and very large models, meaning this is not a practical solution for most research settings and users. There are also diffusion-based audio models such as AudioLDM that solve the problem of resources by working on compressed audio and also support textual guidance. This model also relies on many pretrained components(e.g., CLAP, T5), limiting its reproducibility. All of these works show the battle between controllability, efficiency, and quality, and highlight the need for an accessible, understandable, and lightweight approach to audio generation.

4 Proposed Solution

To address the challenges of computational efficiency and explicit controllability highlighted in Section 3, we propose a **Genre-Conditioned Latent Diffusion Model (GC-LDM)**. Unlike autoregressive models that sample raw audio sequentially (e.g., WaveNet (1), Jukebox (2)), our approach treats audio generation as an image generation task on Mel-spectrograms, utilizing a compressed latent space to minimize resource usage. This architecture effectively disentangles the generation of high-fidelity textures from global structural coherence.

4.1 Justification of Methodology

Our design choices are directly informed by the limitations observed in prior works. While autoregressive models like **Jukebox (2)** offer high fidelity, their sequential sampling process introduces prohibitive latency. Conversely, GAN-based approaches like **MuseGAN (7)** offer speed but historically suffer from training instability and mode collapse.

Therefore, we select a **Latent Diffusion** architecture as the optimal compromise, offering the stable, high-quality synthesis of diffusion models without the computational cost of pixel-space generation. Furthermore, to address the ambiguity and resource overhead of text-conditioned systems (e.g., **AudioLDM (3)**, **MusicLM (8)**), we strictly limit our conditioning to **explicit genre embeddings**. This “Goldilocks” approach—balancing the efficiency of latent spaces with the precision of categorical control—allows us to achieve our goal of accessible, controllable music generation without the complexity of hybrid architectures like **MeLoDy (6)**.

4.2 Data Representation and Compression

Raw audio is high-dimensional and computationally expensive to model directly. We utilize **Mel-spectrograms** as our intermediate representation, as they effectively capture both the harmonic structure and timbral characteristics of music in a 2D format.

To ensure computational feasibility within an academic setting, we employ a **Perceptual Compression Model (VAE)**. A pre-trained Variational Autoencoder (VAE) compresses the Mel-spectrogram x into a lower-dimensional latent representation $z = \mathcal{E}(x)$. By training the diffusion model on these compressed latents rather than pixel-space spectrograms, we significantly reduce the memory and computational cost of both training and inference.

4.3 Model Architecture

Our framework replaces the massive text encoders found in recent works with a lightweight, domain-specific conditioning mechanism. The architecture consists of two primary components:

- **Conditional U-Net Backbone:** The core generative model is a U-Net equipped with cross-attention layers. It learns to reverse a forward diffusion process by iteratively denoising the latent vectors z_t to recover the original signal z_0 .
- **Explicit Genre Embeddings:** Instead of using large language models (LLMs) to interpret vague text prompts, we utilize a learnable embedding layer that maps discrete genre labels (e.g., “Jazz”, “Rock”) to continuous vector representations. These embeddings are injected into the U-Net via cross-attention, providing the model with explicit, unambiguous stylistic constraints without the overhead of semantic text processing.

4.4 Training and Sampling Strategy

The model will be trained on the **GTZAN dataset**, a standard benchmark for genre classification containing 1,000 audio tracks across 10 distinct genres. This selection validates our goal of achieving high-quality results on accessible, non-proprietary datasets.

We will implement **Classifier-Free Guidance** during the sampling process. During training, genre labels are randomly dropped with a fixed probability (e.g., 10%), teaching the model to generate both conditionally and unconditionally. During inference, this allows us to scale the influence of the

genre embedding, giving the user precise control over how strongly the generated audio adheres to the target genre traits.

4.5 Audio Reconstruction

Once the diffusion model generates a clean latent vector z , it is decoded back into a Mel-spectrogram by the VAE decoder $\mathcal{D}(z)$. Finally, to recover the phase information and generate listenable audio, we employ a pre-trained **HiFi-GAN vocoder (9)**, which converts the generated spectrogram into high-fidelity raw audio waveforms. This pipeline ensures that our lightweight generative process still results in professional-quality acoustic output.

5 Novelty and significance

5.1 Novelty

The primary novelty of this work lies in the development of a genre-conditioned deep generative framework that balances fine-grained control with computational efficiency. While large-scale architectures have successfully demonstrated text-to-music synthesis, these systems typically rely on massive proprietary datasets and prohibitive computational resources. Furthermore, prior research often entangles genre control with complex textual prompts or latent representations that are difficult to isolate. Our approach proposes a lightweight, explicit conditioning mechanism that makes controllable music generation accessible within resource-constrained academic environments.

5.2 Significance

This research addresses a critical gap in the democratization of generative audio tools. By enabling explicit genre control without the need for industrial-scale infrastructure, the proposed system offers a practical solution for creators requiring specific musical backdrops in real-time or low-resource settings. From a technical perspective, this project validates the efficacy of deep generative models in capturing high-level musical structure from low-dimensional categorical constraints, contributing to the broader effort of making efficient, controllable generative AI widely accessible.

6 Tentative Timeline

The timeline for the proposed solution is given in Table 1.

Table 1: Tentative Timeline for the proposed solution

Task	Timeline
Research topics and literature survey	1 week
Project proposal report	2 days
Research DGM methodologies	1 week
Data preprocessing and representation	1 week
Mid term Report	2 days
Model design and Training	2 weeks
Music Generation and Evaluation	2 weeks
Report Writing and Documentation	3 days

References

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